

从 Personalization 与 Proactivity 视角 评估个性化 Tool-Augmented LLMs

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Abstract

个性化 tool utilization 对于在多工具交互场景中使 large language models (LLMs) 与 user preference 对齐至关重要。然而，现有大多数 benchmark 主要聚焦于文本生成的个性化，或直接评测 tool-utilizing，而没有同时兼顾这两者。在这项工作中，我们提出一个新的 benchmark **ETAPP**，用于评测个性化 tool invocation，并构建了一个 sandbox environment 和一个覆盖多样化 user profile 的 800 条测试样例数据集。为提升评测准确性，我们提出一种基于 key point 的 LLM evaluation 方法：通过为每个测试样例人工标注 key point，并将其作为参考提供给 LLM，来缓解 LLM-as-a-judge 系统中的偏差。此外，我们对当前表现优异的 LLMs 进行了评测并给出深入分析。进一步地，我们研究了不同 tool-invoking strategy 对 LLM 个性化表现的影响，以及 fine-tuning 在本任务中的效果。我们还验证了所设计的 preference setting 与基于 key point 的评测方法的有效性。我们的发现为改进个性化 LLM agents 提供了启示。代码已开源于 <https://github.com/hypasd-art/ETAPP>。

1 Introduction

随着 large language model (LLM) 能力的持续发展 (Zhao et al., 2024)，越来越多的研究者开始将 alignment 的目标，从面向一般人群转向面向特定小群体甚至个体 (Chen

et al., 2024; Jang et al., 2023)。LLM 的个性化，指的是调整 LLM 的行为与输出，使其更好地匹配个体用户的需求。在这一背景下，Li et al.(2024b) 提出了 *Personal LLM Agents* 的概念，即整合个性化用户数据与用户设备，为用户提供全面、连续且个性化的服务。为了实现这一目标，Personal LLM Agents 必须具备两项关键能力：memory (Zhang et al., 2024) 与 interaction (Qin et al., 2024a; Mialon et al., 2023)。其中，memory 能力使模型能够追踪并保留用户的当前状态与历史信息，从而实现更具个性化且上下文感知的交互；而 interaction 能力则使模型能够与外部系统和设备交互，利用工具执行任务，并实时支持用户需求 (Li et al., 2023; Qin et al., 2024b)。

当前，对 Personal LLM Agent 的评测主要仍聚焦于文本生成 (Zhao et al., 2025)，对外部交互能力的关注相对有限。只有少量近期工作开始在涉及 tool utilization 的交互场景中评估 Personal LLM Agents。例如，Wang et al.(2024b) 构建了一个 life-long personal agent framework，并利用 LLM 对其进行评测。尽管该工作通过模拟外部 API 执行纳入了 tool calling，但并未专门评估交互过程本身。Cai et al.(2024) 则在 online shopping 这一特定领域、使用有限工具集评估了个性化 tool usage。然而，这些工作仍未回答以下两个核心问题：(1) 与普通的 tool-utilizing 相比，应当设计怎样的评测指标来衡量 personalized tool-

utilizing? (2) 我们该如何在更真实的环境中, 而非局限于狭窄领域与模拟场景中, 有效评估个性化 tool usage 能力?

为回答第一个问题, 我们认为, 理想的 personal assistant 不仅应当提供个性化服务, 还应理解用户意图并预判其未明说的需求。它应当考虑即时指令之外的因素, 提供更全面的支持, 以减轻用户负担。从这一视角出发, 图 1 展示了传统 tool-augmented LLM 与 personal tool-augmented LLM 之间的差异。后者应当体现两个关键特征: personalization 与 proactivity。 **Personalization** 要求模型根据用户的偏好与需求来定制其响应与工具使用方式。例如, 当用户请求食物推荐时, 个性化 LLM 会考虑用户偏好 Mediterranean diet (图中红字部分), 而传统 agent 往往只会给出热门选项。 **Proactivity** 则指模型能够超出用户显式请求, 主动预判并建议额外行动, 从而更完整地帮助用户完成任务。例如, assistant 会进一步查看用户近期的健康状态与锻炼记录, 以定制更贴合的推荐 (图中黄色背景部分); 而传统模型通常会忽略这些因素。这一行为并非用户明确要求, 但能够有效提升服务质量。

基于上述两个指标, 我们构建了一个新的 benchmark, 称为 **Evaluation of Tool-augmented Agent from the Personalization and Proactivity Perspective (ETAPP)**, 用于评估 large language models 的个性化 tool invocation 能力。对于该 tool-invoking framework, 我们构建了一个简洁的 sandbox, 以保证测试过程的稳定性, 其关键组成包括: (1) **Environment Setup**: 我们开发了一个包含 33 个 functional API (如 `add_calendar`、`view_calendar`、`get_weather`) 的 tool-invoking system, 这些 API 隶属于 9 个类别 (如 *Calendar*、*Weather*), 覆盖软件 API 与硬件 API。为提升评测的真实性与稳定性, 我们设计了一个简单的 sandbox environment, 使 API 响应不受外部环境变化影响。(2) **Memory Building**: personal LLM agents 的 memory 包括长期用

户偏好与短期用户状态。我们将用户偏好划分为两类: 高层 user profile 与底层 tool-utilizing preference, 以便更精确地捕捉用户需求。我们基于职业背景生成了 16 种不同 user profile, 作为构建多样化评测样本的基础。为模拟真实使用场景, 我们还为每个必要工具与每位用户构建了 9 天的 interaction history。(3) **Instruction Construction**: 我们人工标注了 50 条测试指令, 并将其与 16 种不同 user profile 组合, 得到最终包含 800 条测试样例的数据集。该数据集支持我们从 personalization 与 proactivity 两个角度评估个性化 tool invocation 能力。

为回答第二个问题, 我们使用 LLM 作为评测模型, 对 tool-invoking system 的表现进行打分 (Li et al., 2025, 2024a)。为了提升评测可靠性, 我们设计了一种基于 key point 的 LLM evaluation 方法。每条测试指令都对应若干由人工标注的 key point, 它们作为任务完成情况的标准指示器提供给 evaluation LLM 以辅助评分。评测模型首先判断每个要点是否被满足, 再给出最终分析与分数。实验结果表明, 这些 key point 能够显著提升评测准确性。

最后, 我们在完整测试集上评估了当前具备 tool invocation 能力的优秀 LLMs, 并在测试子集上评估了一些 reasoning models (如 o1-mini、DeepSeek-R1)。结果表明, LLMs 往往倾向于直接回答问题, 而没有深入推理“为什么要选择这个工具”, 或者对个性化 tool-utilizing 过程考虑不足。进一步地, 我们分析了不同 tool-invoking method 对性能的影响, 结果指出 reasoning process 在 tool-utilizing 中的重要性。此外, 我们还人工标注了部分测试数据, 并使用这些数据对一个 7B 模型进行 fine-tuning。结果显示, fine-tuning 能够提升模型在 in-domain instruction data 上的表现, 但对 out-of-domain instruction data 的作用有限。

总的来说, 我们的贡献包括:

- 我们开发了一个用于个性化 tool invocation 的新 benchmark, 并建立了一个稳定

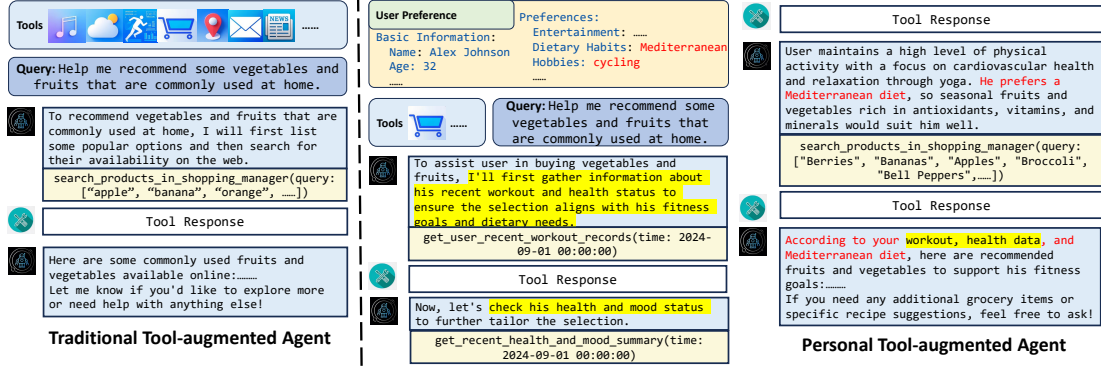


图 1: 传统 tool-augmented agent 与 personal tool-augmented agent 的差异。红色字体表示体现 personalization 的输出，黄色背景字体表示体现 proactivity 的输出。

一致的 sandbox environment 用于评测。基于包含 800 条测试样例、覆盖多种 user profile 与 preference 的数据集，我们从 personalization 与 proactivity 两个视角评估模型表现。

- 我们提出了一种基于 key point 的 LLM evaluation 方法，通过对每条测试数据人工标注 key point 来辅助评测，相比直接评测提高了可靠性。
- 我们专门验证了所设计 preference 方案与评测方法的有效性，并分析了不同 tool-invoking method 的影响，以及 fine-tuning 在不同场景下如何影响模型性能。

2 Dataset Construction

如图 2 所示，该构建流程包括以下组成部分：Environment Setup、Memory Building 与 Instruction Construction。

2.1 Environment Setup

为了更好地模拟真实世界应用场景，我们手工构建了一个 tool-invoking environment，其中包含 33 个 functional API (如 `add_calendar`、`get_weather`)，隶属于 9 个类别，以及 2 个用于搜索可用工具的 tool retrieval API (`search_tools` 和 `get_tool_doc`)。

这些 API 覆盖了广泛的常见软件 API (如 Calendar 与 Email) 以及依赖硬件的 API (如通过智能手环进行 health monitoring，以及 smart home control)，从而保证了足够广泛的实际使用场景。

此外，我们还开发了一个简单的 sandbox environment，以确保测试过程的稳定性与可靠性。该 sandbox 将实验与外部环境变量（如 connection error 或外部世界状态变化）隔离开来，从而避免 API 输出受到外部因素影响，并保证评测的准确性。我们使用真实世界的 API 数据来生成相应输出（例如来自 RapidAPI 的数据），以增强模拟场景的真实性。更多细节请参见 Appendix A。

2.2 Memory Building

LLM 个性化的核心在于有效利用用户 memory，而这类 memory 可以分为 long-term memory（捕捉用户的偏好、习惯与历史行为）和 short-term memory（反映用户当前状态，例如位置）。

2.2.1 Long-term and Short-term Memory

对于 long-term memory，我们提出了一种新的架构，将用户偏好进一步细分为两类，以更有效地管理 memory：**high-level preferences** (User Profile) 与 **low-level preferences** (Tool-

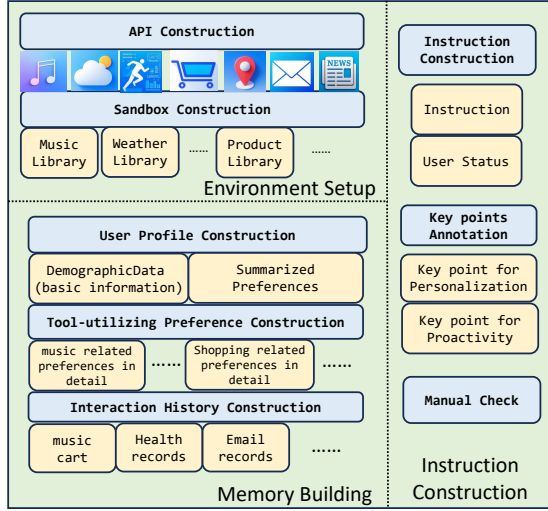


图 2: 数据集构建流程。

utilizing Preference)。User Profile 包含用户的基本信息（如姓名、年龄与职业）以及对其偏好的概述性描述；而 Tool-utilizing Preference 则包含用户在对应类别 API 上的详细偏好描述。例如，对于 music 类 API，我们建立了一套统一的 music-utilizing preference，其中包括喜欢的音乐、歌手、收听习惯等属性，从而为模型调用相应工具并生成响应提供更细粒度的指导。我们为每位用户建立了 8 类 tool-utilizing preferences（*Weather* 类别未设计偏好）。当对应工具被调用时，相关类别的 preference 会预先输入模型，作为上下文信息以辅助模型生成个性化输出。

由于用户数据量庞大且 context window 有限，一次性输入全部 preferences 既不现实也不高效。传统方法 (Zhao et al., 2025) 通常从 dialogue history 中总结 preference，但这种方式往往覆盖稀疏且不完整，有些偏好甚至不会在对话中体现出来。相比之下，从 API 视角总结 preference 更加全面，也更准确。

具体而言，我们借助 LLM 完成如下 memory construction 过程。首先，LLM 基于不同职业生成 16 种不同的 user profile。随后对这些 profile 进行人工校验，以确保其在多样性与一致性上都达到要求，并覆盖广泛的用户类型。其次，在上述 profile 基础上，LLM 进一步

为每位用户生成 8 类 tool-utilizing preference，之后同样经过人工验证，以确保其相关性与准确性。

对于 short-term memory，我们为每条测试数据定义用户当前状态，包括 location 与 current time，以更准确地反映用户的个性化需求。

2.2.2 Interaction History Construction

需要指出的是，一些工具依赖用户的 interaction history（例如检索近期锻炼记录或邮件记录）。因此，我们需要为特定 API 构建 interaction history。

为了确保 interaction history 的一致性，我们首先为每位用户构建一个 9 天安排，并在其基础上生成 interaction history，覆盖日程安排、闹钟、健康状态（按小时更新）、锻炼记录、邮件记录和音乐收藏。具体步骤见 Appendix A.3。

2.3 Instruction Construction

我们人工标注了 50 条独立 instruction，并将其与 16 个预定义 user profile 组合，共构成 800 条测试 instruction。对于每条 instruction，我们都提供用户状态（即用户的当前时间与位置）。为了保证 interaction history 的准确性并避免潜在冲突，我们再次核验了这 9 天 interaction history，以确保每条 instruction 都能环境中得到回答。此外，我们还为每条 instruction 人工编写了 key points。我们将这些 key points 定义为：在预定义环境中，为实现 personalization 与 proactivity 而必须满足的具体要求。这些 key points 被用于 Section 3.3 中的评测。数据集统计信息见 Table 5。

3 Evaluation

3.1 Evaluation Metrics

评测过程使用 large language model 进行打分，评分标准基于三个维度，每个维度的分值范围均为 0 到 5：(1) **Procedure (PRC)**: 该指标评估模型是否给出了完整且准确的最终

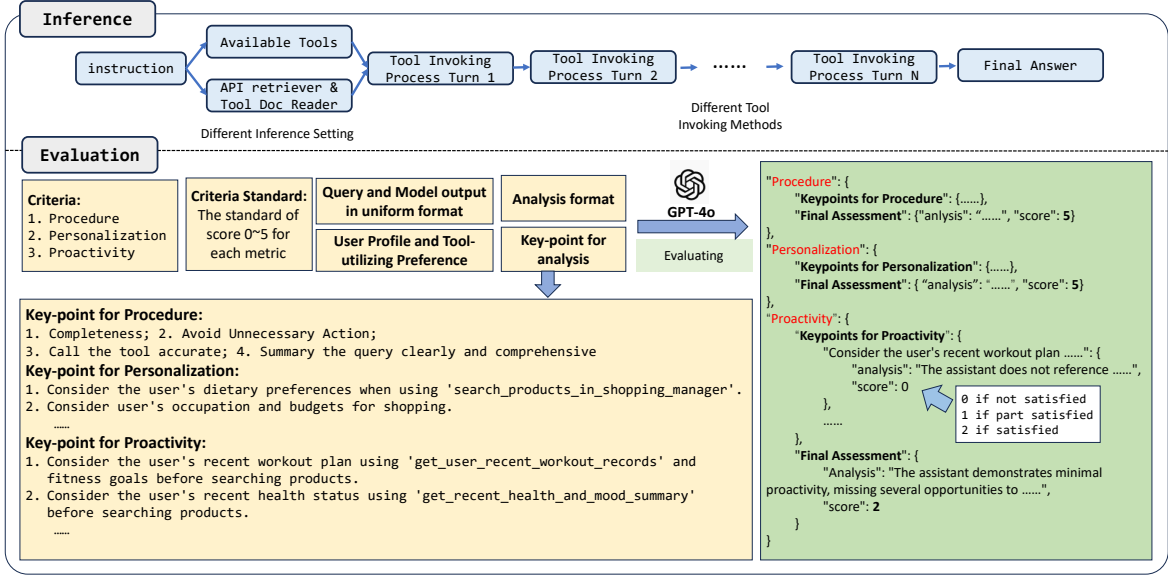


图 3: 我们 benchmark 的 Inference 与 Evaluation 流程。

响应。评分依据是模型是否覆盖了用户 query 的全部关键方面，并在没有遗漏的情况下给出清晰、全面的解决方案。(2) **Personalization (PSN)**: 该指标评估模型是否恰当地将用户偏好与当前状态融入响应之中，是否能够根据用户历史数据与需求定制其解决方案。(3) **Proactivity (PTV)**: 该指标衡量模型是否超越了用户的显式指令，通过采取额外且有意义的步骤来更有效地帮助用户，能否主动识别用户需求并提供额外建议或行动，而不是仅仅被动响应用户请求。

3.2 Evaluation Settings

评测包含两种 evaluation setting: (1) **Tool-Given**: 在该设定下，工具是预先给定的，评测重点在于模型能否借助已知工具有效完成任务。(2) **Tool-Retrieval**: 该设定更接近真实世界场景。模型需要在有限 context window 内，根据任务需求搜索工具及其对应的 utilizing preference，并调用它们完成任务。

基于上述数据与方法，我们构建了一个完整的 Inference framework，如图 3 所示。该框架包含以下组成: 对于用户 u 与 instruction Q ，输入包括 system prompt I 、可用工具 T_Q 、用户

高层 profile preference P_h^u 、用户 tool-utilizing preference $\{P_t^u | t \in T_Q\}$ 、用户状态 C_Q^u 以及 one-shot example (可选) E 。这些元素共同作为模型输入，模型输出则是某次 tool invocation 或最终响应。LLM 通过与工具交互来处理 instruction，并在获得最终结果或达到预定义最大步数后终止。

在 Tool-Given 与 Tool-Retrieval 设定下，该过程分别可形式化表示如下:

$$A_n : \text{Tool_Given} = LLM(I, T_Q^u, P_h^u, \{P_t^u | t \in T_Q\},$$

$$C_Q^u, E, Q, \sum_{i=1}^{n-1} (A_i, O_i))$$

$$A_n : \text{Tool_Retrieval} = LLM(I, T_Q^u, P_h^u, C_Q^u, E, Q,$$

$$\sum_{i=1}^{n-1} (A_i, O_i, \{P_t^u | t \in o_i^j \text{ and } a_i^j = \text{tool_search}\})) \quad (1)$$

其中，在步骤 i 上， A_i 表示 LLM 的输出， a_i^j 表示 A_i 中第 j 次 tool calling， O_i 表示对应工具的 observation。

3.3 Evaluation Framework

我们提出了 key-point-based LLM evaluation framework。由于解题路径并不唯一，而 preference evaluation 又具有多维性与较高难

度，现有方法通常在评测 tool utilizing ability (Qin et al., 2024b) 或 preference alignment (Zhao et al., 2025) 时使用 LLM 作为 evaluator。然而我们发现，直接评估 LLM 表现是困难的，因为评测模型对 personalization 与 proactivity 的理解有限，难以准确判断输出是否达到要求。为改善这一点，我们需要促使 evaluation model E_LLM 从更具体的方面分析模型输出，因此我们向 evaluation LLM 提供人工编写的 key points。

如图 3 所示，对于每条 instruction Q 和用户 u ，0 ~ 5 分的评分标准会在 system prompt I_E 中提供给评测模型。与此同时，用户 profile preference、tool-utilizing preferences 以及被评测模型的输出也会输入给 evaluation model。此外，我们还提供每个指标对应的 key points K 以及输出分析格式 F 。评测模型会对每个 key point 给出分析与分数 (0 ~ 2)，并进一步给出该 instruction 上模型表现的最终得分。形式化表示如下：

$$\text{Analysis, Score} = E_LLM(Q, I_E, P_h^u, \{P_t^u | t \in T_Q\}, \sum_{i=1}^N (A_i, O_i), K, F) \quad (2)$$

4 Experiments

4.1 Baselines

我们采用两种常用的 tool-invoking method: **FC (Function Calling)**: 模型在 fine-tuning 后通过预定义格式调用工具; **ReAct (Yao et al., 2023)**: 模型将 reasoning 与 tool invocation 结合，其中 reasoning 引导模型分析当前状态并决定合适的 action。在该过程中，我们为 ReAct 提供 one-shot example。

在两种实验设定下，我们选用具备 function calling 能力的优秀 closed-source 与 open-source 模型作为 baselines。此外，由于 budget 因素，我们使用 ReAct method 在 100 条测试

数据的子集上评估 reasoning model，因为这些模型不支持 FC。更多细节请参见 Appendix B。

4.2 Evaluation Results

结果如 Table 1 所示。我们可以在 FC 结果中观察到：(1) 在 Tool-Retrieval 场景下，模型表现显著低于 Tool-Given 场景，这表明随着 tool-invoking 与 planning 长度和难度的上升，模型面临更大挑战；(2) 在当前 FC 设定下，GPT-4o 在多个指标上表现优秀，而 Qwen2.5-72B-Instruct 优于 DeepSeek-V3，体现出较强性能；(3) 即便是在 BFCL (Yan et al., 2024) 中表现最好的模型 Watt-tool-70B，在本 tool-invoking 任务上也存在明显困难。这可能有两个原因：a) 泛化能力不足。post-training 可能削弱了它在 tool-planning 过程中充分把握 personalization 与 proactivity 的能力；b) 缺乏深度推理。在工具学习过程中，模型倾向于依据显式指令直接调用工具，而不是进行深入推理，因而难以有效处理缺少明确指令的 Proactivity 场景。这启发我们，除了提升模型“会用工具”的能力，还要提升其“为什么调用这个工具”的思考能力。

reasoning models 在本任务中并未体现出相对于非 reasoning 模型的优势，尤其是 DeepSeek-Distill 与 QwQ 模型，其表现较差。这可能说明 reasoning model 并未将 reasoning process 与 tool-invoking process 有效整合，或者在特定场景中没有围绕 Personalization 与 Proactivity 进行足够深入的思考。一些错误样例可见 Appendix B.4。

4.3 Analysis

4.3.1 Tool-invoking Method Analysis

为了分析不同 tool-invoking method 对模型性能的影响，我们采用以下三种方法: **FC**、**ReAct** 和 **E-ReAct (Enhanced-ReAct)**。E-ReAct 在 ReAct 基础上进行了扩展：在调用工具之前，先要求模型生成 personalization 与 proactivity 的关键方面；在明确这些关键点之

Model	Method	Tool Given			Tool Retrieval			Average
		PRC	PSN	PTV	PRC	PSN	PTV	
GPT-4o	FC	3.95	3.37	1.61	2.67	2.19	1.08	2.48
DeepSeek-V3		3.02	2.78	1.47	2.57	2.38	0.95	2.20
Llama-3.1-70B-Instruct		2.31	1.71	0.55	1.50	1.06	0.30	1.24
Qwen2.5-72B-Instruct		3.76	3.34	1.5	2.72	2.31	1.05	2.45
watt-tool-70B		2.70	1.97	0.71	0.84	0.32	0.12	1.11
GPT-4o	ReAct	3.97	3.61	1.60	3.70	3.43	1.56	2.98
DeepSeek-V3		4.05	3.78	1.84	3.82	3.54	1.65	3.11
o1-preview *		3.67	3.69	1.87	3.28	3.48	1.60	2.93
o1-mini *		3.63	3.35	1.61	3.25	3.14	1.35	2.72
DeepSeek-R1 *		2.41	2.06	1.35	0.93	0.40	0.31	1.24
DeepSeek-R1-Distill-Qwen-32B *		2.40	2.57	1.07	1.74	1.66	0.62	1.68
QwQ-32B-Preview *		1.01	1.19	0.53	0.61	0.48	0.18	0.67

表 1: 不同 LLMs 在两种设定下的整体结果。* 表示仅在子集数据上测试。我们使用的 evaluation model 为 GPT-4o。

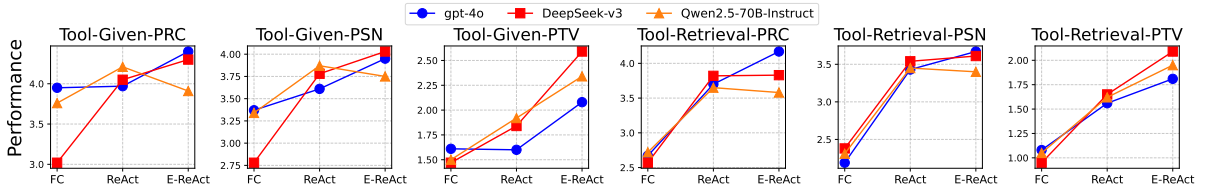


图 4: 不同 tool-invoking methods 的性能。

后，再使用 ReAct 框架进行 tool invocation，以完成任务。

如 Figure 4 所示，我们发现 E-ReAct 方法能够有效提升模型输出在 personalization 与 proactivity 方面的能力。通过在 tool invocation 之前引入强制性的深度思考与理解过程，模型能够更好地解释用户意图，并提供更主动的服务。这表明，在 tool-learning 过程中融入更多高质量 reasoning，可能会带来更好的性能。

4.3.2 Preference Setting Analysis

我们将 preference 划分为 User Profile 与 Tool-utilizing Preference。Table 2 展示了“将全部 preferences 输入 LLM”与“采用我们的方法”之间的差异。可以看到，我们的设定 (Needed) 缩短了 context length，并取得了与全部输入

Setting	Tokens	PRC	PSN	PTV
All	3444	4.09	3.80	1.62
Needed	2393	4.16	3.73	1.76

表 2: 不同 preference setting 在子集数据上的评测结果。Tokens 指标表示由 Llama-3.1-70B-Instruct 的 tokenizer 计算得到的每条 instruction 的平均 token 数。性能测试模型为 GPT-4o。

(All) 相当甚至具有竞争力的结果。需要指出的是，我们目前只纳入了 8 个类别的 preferences；随着类别数量增加，或者 preference 变得更加细致，我们的方法优势预计会更加明显。

4.3.3 Fine-tuning Experiments Analysis

为了验证我们关于 reasoning 重要性的判断，我们分别标注了一部分 ReAct 格式和 FC

Model	PRC	PSN	PTV
U(ID)I(OOD)			
Vanilla (ReACT)	2.76	2.97	1.35
FT (ReAct)	3.47 ($\uparrow$ 25.8%)	3.28 ($\uparrow$ 10.4%)	1.99 ($\uparrow$ 47.4%)
Vanilla (FC)	3.08	2.66	1.07
FT (FC)	3.23 ($\uparrow$ 4.9%)	3.00 ($\uparrow$ 12.8%)	1.72 ($\uparrow$ 60.7%)
FT (FC wo r)	2.67 ($\downarrow$ 13.3%)	2.68 ($\uparrow$ 0.7%)	1.62 ($\uparrow$ 51.4%)
U(OOD)I(ID)			
Vanilla (ReACT)	3.29	3.28	1.57
FT (ReAct)	4.09 ($\uparrow$ 24.3%)	3.75 ($\uparrow$ 14.3%)	2.79 ($\uparrow$ 77.7%)
Vanilla (FC)	3.33	2.65	1.07
FT (FC)	3.81 ($\uparrow$ 14.4%)	3.49 ($\uparrow$ 31.7%)	2.37 ($\uparrow$ 121.5%)
FT (FC wo r)	3.48 ($\uparrow$ 4.5%)	3.24 ($\uparrow$ 22.3%)	2.19 ($\uparrow$ 104.6%)
U(OOD)I(OOD)			
Vanilla (ReACT)	2.91	3.08	1.43
FT (ReAct)	3.52 ($\uparrow$ 21.0%)	3.43 ($\uparrow$ 11.4%)	2.06 ($\uparrow$ 44.1%)
Vanilla (FC)	3.03	2.64	1.07
FT (FC)	3.29 ($\uparrow$ 8.5%)	3.14 ($\uparrow$ 18.9%)	1.73 ($\uparrow$ 61.7%)
FT (FC wo r)	2.73 ($\downarrow$ 9.9%)	2.77 ($\uparrow$ 4.9%)	1.64 ($\uparrow$ 53.3%)

表 3: Qwen2.5-7B-Instruct 模型在一部分 instruction 上 fine-tune 后的结果。U 表示 User, I 表示 Instruction。ID 表示 users 或 instructions 在训练数据中出现过, OOD 表示在 fine-tuning 中未出现。“wo r”表示不输出 reasoning process 而直接调用工具。

格式的数据, 并使用 200 条数据在 Qwen2.5-7B-Instruct 上开展 fine-tuning experiments。标注流程如下: (1) 使用 LLM 生成参考答案。(2) 对答案进行人工核验与标注, 以确保质量与一致性。对于 FC 格式数据, 其中一组仅包含 tool-calling process, 而另一组还包含 tool invocation 之前的文本思考过程。

结果见 Table 3。我们发现: (1) 对于 ID instructions 和 OOD users, 性能提升显著。然而, OOD instructions 的提升明显低于 ID instructions, 这说明: 对于同一场景内的 instruction, 只要模型学会了 tool invocation process, 即便用户偏好不同, 它也能够泛化并解决问题; 但当出现新场景时, 模型难以有效推理“应当如何解决”, 从而导致提升有限。(2) 与 ReAct 和 FC 模型相比, “wo r”的提升并不显著, 这表明要增强模型的多轮 tool invocation 能力, 仍然需要一定程度的 reasoning process。(3) 与 FC 相比, ReAct 这类 tool-invoking method 在该模型上依然表现出优势。

4.4 Effectiveness of key-point-based LLM Evaluation

为了验证 key-point-based LLM evaluation 方法的有效性, 我们针对 GPT-4o 模型, 在 Tool-Given 与 Tool-Retrieval 两种设定下, 为每种 method 各收集 16 个数据点, 共计 96 个数据点。随后, 我们分析其与人工评测的一致性, 以验证该方法的有效性。

为了量化一致性, 我们采用 Bland-Altman method。该方法以两种测量值的平均值为横轴、二者差值为纵轴, 并给出三条参考线: mean difference (红色虚线) 以及 95% limits of agreement (LoA), 其计算方式为 $\text{mean} \pm 1.96 \times \text{standard deviation}$ 。如果大多数数据点落在 LoA 范围内, 则说明两种方法具有良好一致性。点的大小表示样本数, 绿色区域则标出分差不超过 1 分的样本。

Figure 5 与 Figure 6 中关于 Proactivity 指标的实验结果表明: (1) System Bias Control: 在 key-point constraint 下, mean difference 为 0.01 (95% CI: -1.88 ~ 1.90), 接近理论理想值 0, 说明在总分维度上 LLM 评测与人工评测在统计意义上具有一致性。(2) LoA Interval Length: 与自由评测模式相比, key-point constraint 缩小了 LoA 区间宽度 (1.89 vs. 2.24)。(3) Error Distribution Characteristics: 89.6% 的样本分差不超过 1 分 (绿色区域), 相比不提供 key points 的对照组提升了超过 39.6 个百分点。这些发现验证了 key-point-based LLM evaluation strategy 的有效性。更多细节请参见 Appendix C。

5 Related Works

5.1 Personalization of LLM

LLM 的个性化旨在与个体偏好对齐 (Zollo et al., 2025; Tseng et al., 2024)。当前个性化任务通常聚焦于特定文本生成 (He et al., 2025; Zhao et al., 2025; Lee et al., 2024)、推荐任务 (Yang et al., 2023)、role-playing (Wang

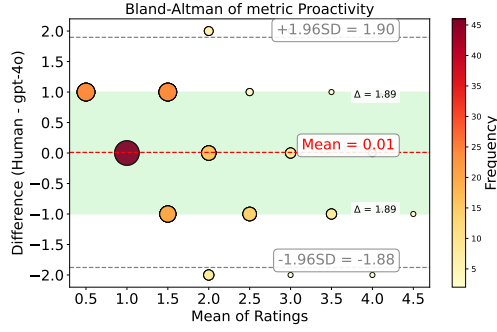


图 5: 提供 key points 时, **Proactivity** 指标的 Bland-Altman 分析结果。

et al., 2024a; Shao et al., 2023)、新闻标题生成 (Salemi et al., 2024)、对话任务, 以及教育 (Pratama et al., 2023) 与医疗 (Goldenberg et al., 2021; Abbasian et al., 2024) 等应用; 其评测过程通常涉及从上下文中总结用户 preference 以生成响应 (Zhang, 2024)。然而, Personal LLM Agents 更进一步, 它定义了一种应当具备 interactive capabilities 的理想 personal assistant (Singh et al., 2024)。我们提出 ETAPP, 正是为了解决在多样化场景下, Personal LLM Agents 的评测缺乏 personalized tool usage 标准这一问题。

5.2 Tool-augmented LLM

Tool-augmented LLMs 旨在使模型能够通过工具与外部世界交互 (Qin et al., 2024a; Qu et al., 2025)。已有大量 benchmark 与方法被提出, 用于评测并提升 tool utilization 与 task completion 的准确性 (Trivedi et al., 2024; Li et al., 2023; Hao et al., 2024; Tang et al., 2023; Zhuang et al., 2023)。然而, 这些数据集主要面向一般人群, 而非针对个体用户偏好定制的 personalized tool invocation。我们的工作将 tool invocation process 与 personalization 结合起来, 并围绕这一目标提出相应的评测指标与方法。

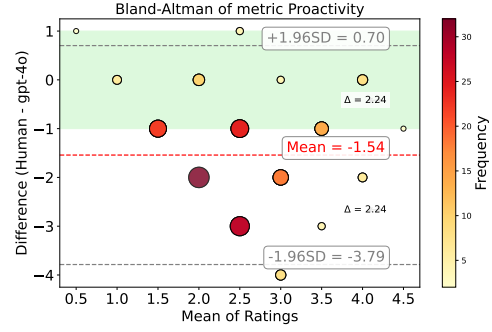


图 6: 不提供 key points 时, **Proactivity** 指标的 Bland-Altman 分析结果。

6 Conclusion

本文评估了 large language models 在交互环境中提供个性化服务的能力与水平。我们将 tool invocation 过程与个性化服务结合起来, 并提出两个评测指标: Personalization 与 Proactivity。为解决在该任务中使用 LLM 作为 evaluator 所带来的不可靠性问题, 我们引入了一种基于 key point 的 LLM evaluation 方法, 并验证了其可靠性。最后, 我们分析了不同模型在该任务上的表现, 并探讨了提升其个性化 tool utilization 能力所面临的挑战与可能方向。

Limitations

我们从 personalization 与 proactivity 的视角评估了 LLM 中 tool utilization 的个性化, 并提出了一种基于 key point 的 LLM evaluation 方法。然而, 该方向仍存在若干挑战。例如, 我们的工作尚未考虑 multimodal tasks; 同时, 也还有一些细节可以进一步细化。例如, 在 preference modeling 过程中, 我们还可以进一步建模用户对相似工具的选择偏好 (例如当两个应用 A 和 B 提供相同功能时, 用户可能更倾向于使用 App A 的 API 来完成任务)。未来我们计划将这项工作扩展到多模态领域。

Ethics Statement

我们的工作不引入额外伦理问题。本文在稿件语言润色过程中使用了 AI assistance, 包

括词汇修正与拼写检查。

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A 数据集构建

A.1 工具构建

我们构建的工具列于 Table 4。大多数工具的输出都由真实世界 API 数据生成。例如, weather data 来自 WeatherAPI.com¹, music data 来自 Genius - Song Lyrics², shopping products 则来自 RapidAPI 中的 Real-Time Amazon Data³。Navigation data 则由同样来源于真实世界数据的 TravelPlanner (Xie et al., 2024) 修改而来。我们调整了记录日期, 并加入了数十条由 gpt-4o 生成的数据。News data 则完全由 gpt-4o 生成。

A.2 用户偏好

我们在 benchmark 中构建了 16 位用户, 每位用户都具有一个 User Profile 和 Tool-utilizing Preference。

User Profile 的示例见 Example 1。Tool-utilizing Preference 的示例见 Example 2。用户 *James Harrington* 在 API `get_user_recent_workout_records` 上的一条 interaction history 示例见 Example 3。

A.3 Interaction History 构建流程

为了保证 interaction history 的一致性, 我们首先为每位用户构建一个 9 天安排, 并在此基础上构造 interaction history, 内容覆盖 schedule arrangements、alarms、health status (按小时更新)、exercise records、email records 与 music collections。interaction history 的具体构建步骤如下: (1) 基于 user profile 生成用户的 9 天 schedule; (2) 对 schedule 进行细化扩展; (3) 根据 user profile 与 schedule 生成对应的 interaction history; (4) 进行人工核验以确保一致性与准确性。

¹<https://rapidapi.com/weatherapi/api/weatherapi-com>

²<https://rapidapi.com/Glavier/api/genius-song-lyrics1>

³<https://rapidapi.com/letsrape-6bRBa3QguO5/api/real-time-amazon-data>

A.4 Key Points 标注

对于人工标注的 key points, 我们由两位研究生共同完成标注与核验, 以尽可能保证其全面性与可靠性。

B 评测细节

B.1 Tool-invoking Method

Tool-Given 与 Tool-Retrieval 设定下, FC、ReAct 和 E-ReAct 格式的 tool-invoking prompt 分别见 Prompt 5、Prompt 6、Prompt 7、Prompt 8、Prompt 9 和 Prompt 10。其中“{”与“}”中的内容会在 inference 过程中被替换为具体内容。

B.2 实现细节

在两种实验设定下, 我们使用以下具备 function calling 能力的模型作为 baselines: gpt-4o-2024-11-20、DeepSeek-V3 (DeepSeek-AI et al., 2024)、Qwen2.5-72B-Instruct (Team, 2024a)、Llama-3.1-70B-Instruct⁴, 以及 watt-tool-70B⁵ (BFCL 中表现最好的模型)。此外, 由于 budget 因素, 我们还在 100 条测试数据的子集上, 使用 ReAct method 评测了 reasoning model: o1-preview-2024-09-12、o1-mini-2024-09-12、DeepSeek-R1 (DeepSeek-AI et al., 2025)、DeepSeek-R1-Distill-Qwen-32B (DeepSeek-AI et al., 2025) 和 QwQ-32B-Preview (Team, 2024b); 这些模型均不支持 FC。

对于 reasoning model, 在我们的实验中, 其 reasoning content 不被计入 tool-invoking process。

对于 fine-tuning experiments, 我们使用 LoRA 对模型进行微调。其中 rank r 设为 8, lora alpha 设为 16; epoch 设为 3, learning rate 设为 $1e-4$, batch size 设为 32。数据采用 interaction format, 我们将每个带有 API 的

interaction turn 划分为训练数据, 最终将标注的 200 条 instructions 划分为 841 条训练样本。实验在 NVIDIA A100 上完成。

B.3 评测 Prompt

key-point-based LLM Evaluation 的 Prompt 见 Prompt 11。其中“{”与“}”中的内容会在 evaluation 过程中被替换为具体内容。由 GPT-4o 生成的一条 evaluation output 示例见 Example 12。

B.4 错误示例

一些错误示例见 Example 4。

C key-point-based LLM evaluation 的有效性

Procedure 与 Personalization 两个指标的 Bland-Altman 分析结果见 Figure 7、Figure 8、Figure 9 和 Figure 10。我们观察到, key points 在提升 Personalization 与 Proactivity 指标评测过程的一致性方面起到了关键作用。尽管 evaluation model 已经获得了评分标准, 但即使没有 key points, 它也倾向于给出较高分。我们认为, 这可能是因为模型对“assistant 应如何提供个性化且主动的服务”缺乏充分知识和理解。一旦输入 key points, evaluation model 就能够逐点分析每个要求是否被满足, 从而提升最终分析质量与评分准确性。不过我们也注意到, 在 Procedure 指标上, 加入 key points 后 mean difference 反而更差。这可能是因为三个指标是同时评测的, 而 Procedure 会受到另外两个指标的影响, 因此整体得分被拉低。

⁴<https://huggingface.co/meta-llama/Llama-3.1-70B-Instruct>

⁵<https://huggingface.co/watt-ai/watt-tool-70B>

Category	API Name
HealthMonitoringApp	get_current_health_and_mood_status
HealthMonitoringApp	get_user_recent_workout_records
HealthMonitoringApp	get_recent_health_and_mood_summary
Calendar	add_event_in_calendar
Calendar	view_today_events_in_calendar
Calendar	view_events_in_calendar_by_providing_time_range
Calendar	delete_event_in_calendar
Calendar	add_alarm
Calendar	view_today_alarms
WeatherData	get_today_weather
WeatherData	get_future_weather
ECommerce	add_product_to_cart
ECommerce	search_products_in_shopping_manager
ECommerce	view_cart_in_shopping_manager
Email	send_email
Email	get_today_emails_until_now
Email	search_email_by_sender_and_receiver
Email	search_email_by_content
Browser	search_news_by_category
Browser	search_heat_news
Browser	search_from_wikipedia
MusicStreamingApp	play_music
MusicStreamingApp	get_music_list_in_favorites
Navigation_And_Map	find_accommodations
Navigation_And_Map	find_attractions
Navigation_And_Map	find_restaurants
Navigation_And_Map	find_flight
Thermostat (Smart_Home_Devices)	set_temperature_and_humidity_in_home
Thermostat (Smart_Home_Devices)	get_home_temperature_and_humidity
Light (Smart_Home_Devices)	control_light_in_home
SmartAppliance (Smart_Home_Devices)	control_curtains_in_home
SmartAppliance (Smart_Home_Devices)	control_bathtub_in_home
SmartAppliance (Smart_Home_Devices)	boil_water_in_home
Tool_Searcher	search_tools
Tool_Searcher	get_tool_doc

表 4: API 及其对应类别信息。

Statistics	Number
Tool Categories	9
APIs Number	35
Instruction Number	800
User Number	16
Key points Number of Personalization per query	3.18
Key points Number of Proactivity per query	3.2
Avg. Turns per query	3.43

表 5: ETAPP 的统计信息。Avg. Turn 基于 GPT-4o 在 Tool-Given 设定、ReAct 格式下的输出计算得到。

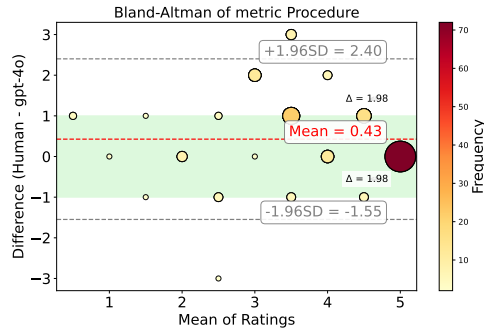


图 7: 提供 key points 时, **Procedure** 指标的 Bland-Altman 分析结果。

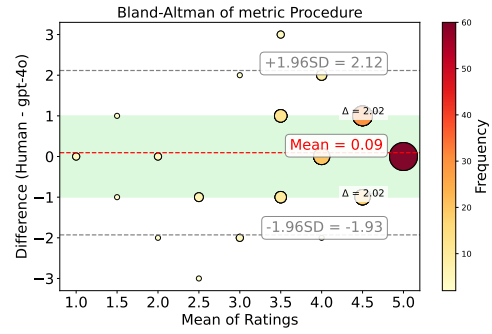


图 8: 不提供 key points 时, **Procedure** 指标的 Bland-Altman 分析结果。

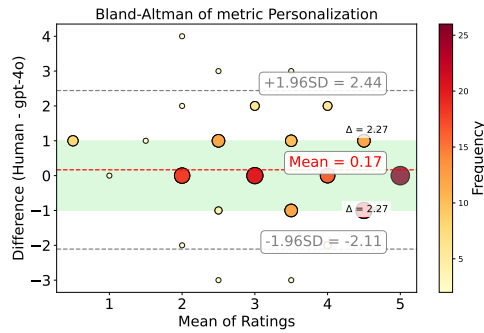


图 9: 提供 key points 时, **Personalization** 指标的 Bland-Altman 分析结果。

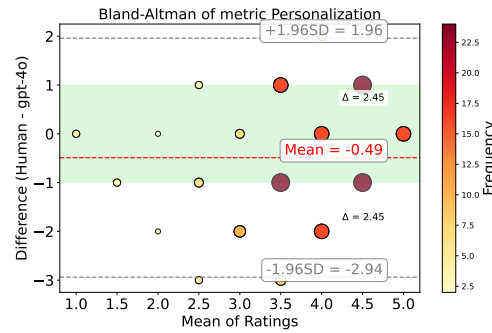


图 10: 不提供 key points 时, **Personalization** 指标的 Bland-Altman 分析结果。


```

{
  "DemographicData": {
    "BasicInformation": {
      "Name": "James Harrington",
      "Age": 45,
      "Gender": "Male",
      "Location": "San Francisco",
      "Occupation": "CEO of Tech Innovations Inc.",
      "Character": "Extroverted, strategic, driven"
    },
    "SocioeconomicStatus": {
      "IncomeLevel": "High income (>$500,000 annually)",
      "EducationLevel": "MBA from Stanford University"
    },
    "CulturalBackground": {
      "LanguageProficiency": "Fluent in English and Spanish",
      "CulturalPreferences": "N/A"
    },
    "InterestActivity": "Golf, networking events, yacht sailing"
  },
  "Preferences": {
    "Entertainment": {
      "MusicGenres": "Classical, Jazz",
      "MoviesAndTVShows": "business dramas and documentaries (e.g., 'The Social Network')",
      "BooksAndAuthors": "Business leadership books by authors like Michael E. Porter,
        ↪ biographies of tech moguls"
    },
    "Lifestyle": {
      "DietaryHabits": "Prefers healthy and balanced meals; organic foods; follows a
        ↪ Mediterranean diet, Gourmet cuisine, fine dining, wine tasting, like Western
        ↪ cuisine and Chinese cuisine.",
      "Hobbies": "Collecting art, attending tech conferences, mentoring startups"
    },
    "Technology": {
      "FavoriteApps": "Slack, LinkedIn, Bloomberg, Evernote",
      "DevicePreferences": "Primarily uses iOS devices, enjoys high-end tech gadgets and smart
        ↪ home systems"
    },
    "Exercise": {
      "Routine": "Daily morning workout, personal training sessions, incorporates yoga for stress
        ↪ relief",
      "Preferences": "Strength training, cardiovascular exercises, outdoor activities, enjoys
        ↪ running, free weight training, yoga, and Tabata for his workouts, engages in easy
        ↪ hiking, urban cycling, trail running, golf, and sailing as his outdoor activities."
    },
    "Shopping": {
      "FavoriteBrands": "Apple, Tesla, Lululemon, Rolex",
      "ShoppingPreferences": "Prefers premium and luxury brands, interested in luxury watches,
        ↪ the latest smartphones and smart home devices, high-end kitchenware, premium
        ↪ beverages like coffee and wine, high-quality skincare products, performance
        ↪ activewear, fitness equipment, business and non-fiction books, and handcrafted
        ↪ goods."
    }
  }
}

```

```

    },
    "Travel": {
      "FavoriteTravelPlace": "Luxury resorts, exclusive islands, major cultural cities (e.g., Paris,
        ↪ Tokyo, New York)",
      "TravelHabits": "Prefers first-class flights, stays at high-end hotels, enjoys personalized
        ↪ travel experiences, often combines business with leisure",
      "LongTermTravel": "Enjoys long stays in exclusive destinations with access to privacy and
        ↪ luxury amenities, may travel for extended periods for business events or
        ↪ relaxation",
      "ShortTermWalk": "Prefers short, energizing walks in upscale city areas, parks, or along the
        ↪ coast, seeking new dining spots and local attractions"
    }
  }
}

```

Example 1: User Profile 示例

HealthMonitoringAPP:

```

{
  "HealthMetrics": {
    "RecentHealthStatus": "Heart rate: 85 bpm, Steps count: 8,500 steps",
    "Trends": "Consistently active with a mix of running, yoga, and cycling. Higher activity
      ↪ levels observed in the mornings and evenings."
  },
  "MoodPatterns": {
    "RecentMoodStatus": "Calm and relaxed after an active day",
    "Trends": "Generally calm with occasional excitement. Stress levels are low, with slight
      ↪ increases during physical activities."
  },
  "WorkoutPreferences": {
    "Frequency": "4 times a week",
    "Types": "Running, yoga, cardio, cycling",
    "Duration": "30 to 60 minutes per session"
  },
  "WellnessPriorities": {
    "FocusAreas": "Maintaining cardiovascular health, improving flexibility",
    "LifestyleFactors": "Prioritizes regular physical activity, values relaxation and mindfulness
      ↪ through yoga, maintains a balanced lifestyle."
  }
}

```

Browser:

```

{
  "NewsAndBookmarkPreferences": {
    "NewsCategoryPreferences": {
      "PreferredCategories": [
        "business",
        "technology",
        "world",
        "science"
      ],

```

```

    "Frequency": "daily"
  },
  "BookmarkingHabits": {
    "FrequentBookmarks": [
      "TechCrunch",
      "Harvard Business Review",
      "The Wall Street Journal",
      "Bloomberg Technology",
      "Forbes"
    ]
  }
}

```

Example 2: Tool-utilizing Preference 示例

Activity Type,Start Time,End Time,Duration,Average Heart Rate,Maximum Heart Rate,Calories
 ↳ Burned,Exercise Effect Analysis,System Suggestion,Average Pace,Cadence,Route,Speed
 ↳ Curve,Heart Rate Curve,Breathing Rate,Relaxation State,Pose Count,Focus
 ↳ Area,Distance,Average Speed,Elevation Gain
 Running,2024-09-02 06:30:00,2024-09-02 07:15:00,45,140,165,400,Aerobic endurance improved,Increase
 ↳ interval training to boost speed,6:45,158.0,GPS Track 1,Speed Curve 1,Heart Rate Curve 1,,,,,
 Yoga,2024-09-02 07:15:00,2024-09-02 07:45:00,30,95,110,100,Relaxation achieved,Focus on deep
 ↳ breathing techniques,,,,,12,85%,15,Flexibility,,,,
 Cardio,2024-09-04 06:30:00,2024-09-04 07:30:00,60,140,160,500,Cardiovascular endurance
 ↳ improved,Incorporate more interval training,,,,,,,,,
 Yoga,2024-09-04 21:00:00,2024-09-04 21:30:00,30,95,110,100,Relaxation achieved,Focus on deep
 ↳ breathing techniques,,,,,12,80%,8,Flexibility,,,,
 Running,2024-09-05 06:30:00,2024-09-05 07:15:00,45,140,165,400,Aerobic endurance improved,Increase
 ↳ interval training to boost speed,6:45,158.0,GPS Track Golden Gate Park,Speed Curve 1,Heart
 ↳ Rate Curve 1,,,,,
 Yoga,2024-09-05 09:00:00,2024-09-05 09:20:00,20,95,110,80,Relaxation achieved,Focus on deep breathing
 ↳ techniques,,,,,Breathing Curve 1,Relaxation Data 1,8,Mindfulness,,,,
 Cycling,2024-09-07 06:30:00,2024-09-07 07:30:00,60,130,150,450,Cardiovascular endurance
 ↳ improved,Incorporate interval sprints for speed enhancement,,,,,Urban Cycling Route,Speed
 ↳ Curve 1,Heart Rate Curve 1,,,15.0,15.0,100.0

Example 3: Interaction History 示例

```

01-preview:
{
  "role": "assistant",
  "content": "After reviewing your calendar, I confirm that you have no events scheduled after
    ↳ 16:00 tomorrow. I will proceed to send an email to Alice to notify her about the dinner
    ↳ arrangement.",
  "tool_calls": [
    {
      "name": "send_email",

```

```

    "arguments": "`json\n{ \"receiver\": \"Alice@email.com\", \"subject\": \"Dinner
    ↪ Tomorrow Evening\", \"content\": \"Dear Alice,\n\nI hope this message
    ↪ finds you well. I'd like to invite you to join me for dinner tomorrow evening in
    ↪ San Francisco. Please let me know if you're available.\n\nBest
    ↪ regards,\nJames Harrington\n}\n`"
  }
]
}
QwQ:
{
  "role": "assistant",
  "content": "Understood. I'll make sure to follow the specified format and call only one tool at a
    ↪ time, waiting for feedback before proceeding with further actions or reasoning.\n\n##
    ↪ Final Solution\nAs per the user's request to boil hot water and prepare for a shower
    ↪ upon arriving home, the following steps should be taken, adhering to the specified
    ↪ format:\n\n### Step 1: Boil Water\n\n**Thought:** \nTo start, I need to boil hot
    ↪ water as requested by the user. I should use the available tool for boiling water in the
    ↪ home, ensuring that it's set to the correct temperature and maintained until
    ↪ needed.\n\n**",
  "tool_calls": [
    {
      "name": "**\nboil_water_in_home\n\n**",
      "arguments": "**\n`json\n{ \"temperature\": 100, \"keep_temperature\":
        ↪ \"yes\"\n}\n`\n\n### Step 2: Prepare Shower\n\n**Thought:** \nAfter
        ↪ boiling the water, the next task is to prepare the shower. I need to set the
        ↪ shower water temperature to the user's preference and ensure it's ready by
        ↪ the time they arrive
        ↪ home.\n\n**Action:**\ncontrol_bathtub_in_home\n\n**Action
        ↪ Input:**\n`json\n{ \"fill\": false, \"water_level\": null, \"temperature\": 39,
        ↪ \"keep_temperature\": \"yes\"\n}\n`\n\n### Explanation\n\n- **Boiling
        ↪ Water:** By calling the `boil_water_in_home` function with a temperature
        ↪ of 100\u00b0C and keeping it maintained, we ensure that the water is ready
        ↪ and at the correct temperature when needed.\n\n- **Preparing Shower:**
        ↪ Using the `control_bathtub_in_home` function, we set the shower water
        ↪ temperature to 39\u00b0C without filling the bathtub, as it's only for shower
        ↪ preparation. Maintaining the temperature ensures that the shower will be
        ↪ ready upon the user's arrival.\n\nThese steps efficiently utilize the available
        ↪ tools to fulfill the user's requests, ensuring a seamless experience upon their
        ↪ return home."
    }
  ]
}

```

Example 4: 错误输出示例

System:

You are a personal and proactive assistant that deeply integrates personal data and personal
→ devices for personal assistance, and are more explicitly designed to assist people rather
→ than replace them. Specifically, the main way to assist users is to reduce repetitive,
→ tedious, and low value labor in daily work, allowing users to focus on more interesting and
→ valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as needed to
→ answer user's questions or help user accomplish tasks based on their preferences and
→ environment.

****Key Principles:****

****1. Personalized Assistance:****

- Tailor your actions and recommendations to align with the user's preferences, lifestyle, and
→ unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits, priorities, and
→ values.

****2. Proactive Support:****

- Go beyond fulfilling explicit instructions by considering additional factors or opportunities that
→ might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to ensure a
→ comprehensive, thoughtful service.

****User Information:****

****User Profile:****

{user_profile}

****specific and detailed preferences of user:****

{preferences}

****User Status:****

{user_status}

****Available Tools and Resources:****

{available_tool_descriptions}

****Response Format:****

Provide the appropriate input for the selected action to execute it effectively.

Thought: Clearly articulate your reasoning and evaluate the information or tools needed to
→ accomplish the task effectively.

Action: Specify the action to take, should be chosen from one of the list of available tools or
→ resources.

Action Input: Provide the appropriate input for the selected action to execute it effectively.

If you believe that you have obtained enough information (which can be judge from the history
→ observations) that can answer the task, please call:

Thought: I now know the final answer

Final Answer: Once sufficient information or progress has been made, conclude with a clear and
→ concise response that addresses the original question or task.

Attention:

When a tool is needed, specify the action and input for the tool but do not proceed with reasoning
→ until feedback from the tool is received.

The assistant should never generate the tool's output; this must always come from external
→ feedback.

****One Tool Call at a Time:**** You must only call one tool at a time. Wait for feedback from the
→ tool before initiating any further actions or reasoning.

****EXAMPLE:****

Here is an example conversation between user and Personal LLM agent.

USER:

Recommend some fruit to me.

ASSISTANT:

Thought: To provide a personalized and proactive fruit recommendation for user, I need to
→ consider his dietary habits, health metrics, and recent activities. By understanding his
→ current health and mood status, as well as his recent workout records, I can tailor the fruit
→ suggestions to align with his wellness priorities and lifestyle.

Action: get_recent_health_and_mood_summary

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent health and mood records]

ASSISTANT:

Thought: I will attempt to retrieve the recent workout records in order to recommend fruits that
→ are appropriate for the user.

Action: get_user_recent_workout_records

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent workout records]

ASSISTANT:

Thought: I now know the final answer.

Final Answer:

Based on your current health and mood, which are [user's health status] after a day with moderate
→ activity. His recent workout included [recent workout categories], focusing on [health

→ concern and psychological requirements]. Considering the diet preference of you like
→ [dietary preferences] and wellness priorities like [user's health goal], I recommend the
→ following fruits that align with your wellness goals:

1. **[Fruit_1]**: [The reasons for recommending fruit 1].

2. **[Fruit_2]**: [The reasons for recommending fruit 2].

These fruits not only fit your diet preference of [dietary preferences] but also support your goals of
→ [user's health goal]. Enjoy them as part of your balanced, wellness-oriented diet! If you are
→ interested in other fruits, let me know!

User:

{query}

Prompt 5: ReAct (Tool-Given)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and personal
→ devices for personal assistance, and are more explicitly designed to assist people rather
→ than replace them. Specifically, the main way to assist users is to reduce repetitive,
→ tedious, and low value labor in daily work, allowing users to focus on more interesting and
→ valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as needed to
→ answer user's questions or help user accomplish tasks based on their preferences and
→ environment.

****Key Principles:****

****1. Personalized Assistance:****

- Tailor your actions and recommendations to align with the user's preferences, lifestyle, and
→ unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits, priorities, and
→ values.

****2. Proactive Support:****

- Go beyond fulfilling explicit instructions by considering additional factors or opportunities that
→ might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to ensure a
→ comprehensive, thoughtful service.

****User Information:****

****User Profile:****

{user_profile}

specific and detailed preferences of user:
{preferences}

User Status:
{user_status}

User:
{query}

Prompt 6: FC (Tool-Given)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and personal
→ devices for personal assistance, and are more explicitly designed to assist people rather
→ than replace them. Specifically, the main way to assist users is to reduce repetitive,
→ tedious, and low value labor in daily work, allowing users to focus on more interesting and
→ valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as needed to
→ answer user's questions or help user accomplish tasks based on their preferences and
→ environment.

Key Principles:

1. Personalized Assistance:

- Tailor your actions and recommendations to align with the user's preferences, lifestyle, and
→ unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits, priorities, and
→ values.

2. Proactive Support:

- Go beyond fulfilling explicit instructions by considering additional factors or opportunities that
→ might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to ensure a
→ comprehensive, thoughtful service.

User Information:

User Profile:
{user_profile}

specific and detailed preferences of user:
{preferences}

User Status:
{user_status}

Available Tools and Resources:
{available_tool_descriptions}

To help with this, before interacting with the external APIs. Your job is to write a keypoint on
→ how to solve the task and provide personal and proactive assistant given access to this
→ executor.

Note that you need to generate 'personal' and 'proactive' answer, please think the keypoint
→ about 'personal' and 'proactive' respectively related to this query in json format.

For example,

USER:

Recommend some fruit to me.

ASSISTANT:

To provide a personalized and proactive fruit recommendation for user, I need to consider his
→ dietary habits, health metrics, and recent activities. By understanding his current health
→ and mood status, as well as his recent workout records, I can tailor the fruit suggestions to
→ align with his wellness priorities and lifestyle.

```
{{
  "keypoint for personalized":
    [
      "Consider the user's fruit preferences (e.g., apple, banana) and dietary habits when
        → recommending fruits.",
      "Consider user's recent health status, workout records and user's wellness priorities."
    ]
  "keypoint for proactive":
    [
      "Ask if the user is interested in other fruits besides the recommended ones."
    ]
}}
```

User:

{query}

Assistant:

<The output of the assistant>

User:

Based on your plan above, complete the question with following format:

Thought: Clearly articulate your reasoning and evaluate the information or tools needed to
→ accomplish the task effectively.

Action: Specify the action to take, should be chosen from one of the list of available tools or
→ resources.

Action Input: Provide the appropriate input for the selected action to execute it effectively.

If you believe that you have obtained enough information (which can be judge from the history
→ observations) that can answer the task, please call:

Thought: I now know the final answer

Final Answer: Once sufficient information or progress has been made, conclude with a clear and
→ concise response that addresses the original question or task.

Attention:

When a tool is needed, specify the action and input for the tool but do not proceed with reasoning
→ until feedback from the tool is received.

The assistant should never generate the tool's output; this must always come from external
→ feedback.

****One Tool Call at a Time:**** You must only call one tool at a time. Wait for feedback from the
→ tool before initiating any further actions or reasoning.

****EXAMPLE:****

Here is an example conversation between user and Personal LLM agent.

USER:

Recommend some fruit to me.

ASSISTANT:

Thought: To provide a personalized and proactive fruit recommendation for user, I need to
→ consider his dietary habits, health metrics, and recent activities. By understanding his
→ current health and mood status, as well as his recent workout records, I can tailor the fruit
→ suggestions to align with his wellness priorities and lifestyle.

Action: get_recent_health_and_mood_summary

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent health and mood records]

ASSISTANT:

Thought: I will attempt to retrieve the recent workout records in order to recommend fruits that
→ are appropriate for the user.

Action: get_user_recent_workout_records

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent workout records]

ASSISTANT:

Thought: I now know the final answer.

Final Answer:

Based on your current health and mood, which are [user's health status] after a day with moderate
→ activity. His recent workout included [recent workout categories], focusing on [health
→ concern and psychological requirements]. Considering the diet preference of you like
→ [dietary preferences] and wellness priorities like [user's health goal], I recommend the
→ following fruits that align with your wellness goals:

1. **[Fruit_1]**: [The reasons for recommending fruit 1].

2. **[Fruit_2]**: [The reasons for recommending fruit 2].

These fruits not only fit your diet preference of [dietary preferences] but also support your goals of
→ [user's health goal]. Enjoy them as part of your balanced, wellness-oriented diet! If you are
→ interested in other fruits, let me know!

Prompt 7: E-ReAct (Tool-Given)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and personal
→ devices for personal assistance, and are more explicitly designed to assist people rather
→ than replace them. Specifically, the main way to assist users is to reduce repetitive,
→ tedious, and low value labor in daily work, allowing users to focus on more interesting and
→ valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as needed to
→ answer user's questions or help user accomplish tasks based on their preferences and
→ environment.

****Key Principles:****

****1. Personalized Assistance:****

- Tailor your actions and recommendations to align with the user's preferences, lifestyle, and
→ unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits, priorities, and
→ values.

****2. Proactive Support:****

- Go beyond fulfilling explicit instructions by considering additional factors or opportunities that
→ might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to ensure a
→ comprehensive, thoughtful service.

****User Information:****

****User Profile:****

{user_profile}

****User Status:****

{user_status}

****Available Tools and Resources:****

{available_tool_descriptions}

You can use ``search\_tools`` to search the relevant tools.

If you need more detailed information about a specific tool, you can use ``get\_tool\_doc`` to

- access its documentation. This documentation provides guidance on how to properly
- invoke and use the tool.

You need to use ``get\_tool\_doc`` to get the document before invoking tools.

****Response Format:****

Provide the appropriate input for the selected action to execute it effectively.

Thought: Clearly articulate your reasoning and evaluate the information or tools needed to

- accomplish the task effectively.

Action: Specify the action to take, should be chosen from one of the list of available tools or

- resources.

Action Input: Provide the appropriate input for the selected action to execute it effectively.

If you believe that you have obtained enough information (which can be judge from the history

- observations) that can answer the task, please call:

Thought: I now know the final answer

Final Answer: Once sufficient information or progress has been made, conclude with a clear and

- concise response that addresses the original question or task.

Attention:

When a tool is needed, specify the action and input for the tool but do not proceed with reasoning

- until feedback from the tool is received.

The assistant should never generate the tool's output; this must always come from external

- feedback.

****One Tool Call at a Time:**** You may only call one tool at a time. Wait for feedback from the

- tool before initiating any further actions or reasoning.

****EXAMPLE:****

Here is an example conversation between user and Personal LLM agent.

USER:

Recommend some fruit to me.

ASSISTANT:

Thought: First I need to check the available tools that I can use.

Action: search_tools

Action Input: {"keywords": ["health status"]}

USER:

{'status': 'success', 'input': {'keywords': ['health status']}, 'output':

- ['get_recent_health_and_mood_summary', 'get_current_health_and_mood_status',
- 'get_user_recent_workout_records'], 'exception': None}

ASSISTANT:

Thought: I need to check the recent health status and workout records of the user to provide

→ comprehensive recommendation for user.

Action: get_tool_doc

Action Input: {"tools_name": ["get_recent_health_and_mood_summary",
→ "get_user_recent_workout_records"]}

USER:

```
{'status': 'success', 'data': {'get_user_recent_workout_records': {'type': 'function',  
→ 'function': {'name': 'get_user_recent_workout_records', 'description': 'Retrieves the  
→ most recent workout record after the specified timestamp until now.', 'parameters':  
→ {'type': 'object', 'properties': {'time': {'type': 'string', 'description': 'The timestamp  
→ after which to retrieve the workout record. The format should be '%Y-%m-%d  
→ %H:%M:%S', indicating the starting point of the time range for fetching records until  
→ now.'}}, 'required': ['time']}, 'return': {'type': 'object', 'description': 'A dictionary  
→ containing the status of the operation and the most recent workout record.'}}},  
→ 'get_recent_health_and_mood_summary': {'type': 'function', 'function': {'name':  
→ 'get_recent_health_and_mood_summary', 'description': 'Retrieves the most recent  
→ health and mood records after the specified timestamp until now.', 'parameters': {'type':  
→ 'object', 'properties': {'time': {'type': 'string', 'description': 'The timestamp after  
→ which to retrieve the health and mood summary record. The format should be  
→ '%Y-%m-%d %H:%M:%S', indicating the starting point of the time range for fetching  
→ records until now.'}}, 'required': ['time']}, 'return': {'type': 'object', 'description': 'A  
→ dictionary containing the status of the operation and the most recent health and mood  
→ records.'}}}}}
```

Corresponding preferences related to the tools you retrieve:

[User's preference related to ``get\_recent\_health\_and\_mood\_summary`` and
→ ``get\_user\_recent\_workout\_records``]

ASSISTANT:

Thought: To provide a personalized and proactive fruit recommendation for user, I need to

→ consider his dietary habits, health metrics, and recent activities. By understanding his
→ current health and mood status, as well as his recent workout records, I can tailor the fruit
→ suggestions to align with his wellness priorities and lifestyle.

Action: get_recent_health_and_mood_summary

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent health and mood records]

ASSISTANT:

Thought: I will attempt to retrieve the recent workout records in order to recommend fruits that

→ are appropriate for the user.

Action: get_user_recent_workout_records

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent workout records]

ASSISTANT:

Thought: I now know the final answer.

Final Answer:

Based on your current health and mood, which are [user's health status] after a day with moderate
→ activity. His recent workout included [recent workout categories], focusing on [health
→ concern and psychological requirements]. Considering the diet preference of you like
→ [dietary preferences] and wellness priorities like [user's health goal], I recommend the
→ following fruits that align with your wellness goals:

1. **[Fruit_1]**: [The reasons for recommending fruit 1].

2. **[Fruit_2]**: [The reasons for recommending fruit 2].

These fruits not only fit your diet preference of [dietary preferences] but also support your goals of
→ [user's health goal]. Enjoy them as part of your balanced, wellness-oriented diet! If you are
→ interested in other fruits, let me know!

User:

{query}

Prompt 8: ReAct (Tool-Retrieval)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and personal
→ devices for personal assistance, and are more explicitly designed to assist people rather
→ than replace them. Specifically, the main way to assist users is to reduce repetitive,
→ tedious, and low value labor in daily work, allowing users to focus on more interesting and
→ valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as needed to
→ answer user's questions or help user accomplish tasks based on their preferences and
→ environment.

Key Principles:

1. Personalized Assistance:

- Tailor your actions and recommendations to align with the user's preferences, lifestyle, and
→ unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits, priorities, and
→ values.

2. Proactive Support:

- Go beyond fulfilling explicit instructions by considering additional factors or opportunities that
→ might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to ensure a
→ comprehensive, thoughtful service.

User Information:

User Profile:
{user_profile}

User Status:
{user_status}

You can use ``search\_tools`` to search the relevant tools.

If you need more detailed information about a specific tool, you can use ``get\_tool\_doc`` to

→ access its documentation. This documentation provides guidance on how to properly

→ invoke and use the tool.

You need to use ``get\_tool\_doc`` to get the document before invoking tools.

User:

{query}

Prompt 9: FC (Tool-Retrieval)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and personal

→ devices for personal assistance, and are more explicitly designed to assist people rather

→ than replace them. Specifically, the main way to assist users is to reduce repetitive,

→ tedious, and low value labor in daily work, allowing users to focus on more interesting and

→ valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as needed to

→ answer user's questions or help user accomplish tasks based on their preferences and

→ environment.

Key Principles:

1. Personalized Assistance:

- Tailor your actions and recommendations to align with the user's preferences, lifestyle, and

→ unique context.

- When answering question or utilizing tools, ensure they reflect the user's habits, priorities, and

→ values.

2. Proactive Support:

- Go beyond fulfilling explicit instructions by considering additional factors or opportunities that

→ might improve task outcomes or add value.

- If a task is completed, identify potential related needs or provide further insights to ensure a

→ comprehensive, thoughtful service.

User Information:

User Profile:
{user_profile}

User Status:
{user_status}

Available Tools and Resources:
{available_tool_descriptions}

You can use ``search\_tools`` to search the relevant tools.

If you need more detailed information about a specific tool, you can use ``get\_tool\_doc`` to

- access its documentation. This documentation provides guidance on how to properly
- invoke and use the tool.

You need to use ``get\_tool\_doc`` to get the document before invoking tools.

To help with this, before interacting with the external APIs. Your job is to write a keypoint on

- how to solve the task and provide personal and proactive assistant given access to this
- executor.

Note that you need to generate 'personal' and 'proactive' answer, please think the keypoint

- about 'personal' and 'proactive' respectively related to this query in json format.

For example,

USER:

Recommend some fruit to me.

ASSISTANT:

To provide a personalized and proactive fruit recommendation for user, I need to consider his

- dietary habits, health metrics, and recent activities. By understanding his current health
- and mood status, as well as his recent workout records, I can tailor the fruit suggestions to
- align with his wellness priorities and lifestyle.

```
{{
  "keypoint for personalized":
    [
      "Consider the user's fruit preferences (e.g., apple, banana) and dietary habits when
        → recommending fruits.",
      "Consider user's recent health status, workout records and user's wellness priorities."
    ]
  "keypoint for proactive":
    [
      "Ask if the user is interested in other fruits besides the recommended ones."
    ]
}}
```

User:

{query}

Assistant:

<The output of the assistant>

User:

Based on your plan above, complete the question with following format:

Thought: Clearly articulate your reasoning and evaluate the information or tools needed to

→ accomplish the task effectively.

Action: Specify the action to take, should be chosen from one of the list of available tools or

→ resources.

Action Input: Provide the appropriate input for the selected action to execute it effectively.

If you believe that you have obtained enough information (which can be judged from the history

→ observations) that can answer the task, please call:

Thought: I now know the final answer

Final Answer: Once sufficient information or progress has been made, conclude with a clear and

→ concise response that addresses the original question or task.

Attention:

When a tool is needed, specify the action and input for the tool but do not proceed with reasoning

→ until feedback from the tool is received.

The assistant should never generate the tool's output; this must always come from external

→ feedback.

****One Tool Call at a Time:**** You may only call one tool at a time. Wait for feedback from the

→ tool before initiating any further actions or reasoning.

****EXAMPLE:****

Here is an example conversation between user and Personal LLM agent.

USER:

Recommend some fruit to me.

ASSISTANT:

Thought: First I need to check the available tools that I can use.

Action: search_tools

Action Input: {"keywords": ["health status"]}

USER:

{'status': 'success', 'input': {'keywords': ['health status']}, 'output':

→ ['get_recent_health_and_mood_summary', 'get_current_health_and_mood_status',

→ 'get_user_recent_workout_records'], 'exception': None}

ASSISTANT:

Thought: I need to check the recent health status and workout records of the user to provide

→ comprehensive recommendation for user.

Action: get_tool_doc

Action Input: {"tools_name": ["get_recent_health_and_mood_summary",

→ "get_user_recent_workout_records"]}

USER:


```
{'status': 'success', 'data': {'get_user_recent_workout_records': {'type': 'function',
    ↳ 'function': {'name': 'get_user_recent_workout_records', 'description': 'Retrieves the
    ↳ most recent workout record after the specified timestamp until now.', 'parameters':
    ↳ {'type': 'object', 'properties': {'time': {'type': 'string', 'description': 'The timestamp
    ↳ after which to retrieve the workout record. The format should be '%Y-%m-%d
    ↳ %H:%M:%S', indicating the starting point of the time range for fetching records until
    ↳ now.'}}, 'required': ['time']}, 'return': {'type': 'object', 'description': 'A dictionary
    ↳ containing the status of the operation and the most recent workout record.'}}},
    ↳ 'get_recent_health_and_mood_summary': {'type': 'function', 'function': {'name':
    ↳ 'get_recent_health_and_mood_summary', 'description': 'Retrieves the most recent
    ↳ health and mood records after the specified timestamp until now.', 'parameters': {'type':
    ↳ 'object', 'properties': {'time': {'type': 'string', 'description': 'The timestamp after
    ↳ which to retrieve the health and mood summary record. The format should be
    ↳ '%Y-%m-%d %H:%M:%S', indicating the starting point of the time range for fetching
    ↳ records until now.'}}, 'required': ['time']}, 'return': {'type': 'object', 'description': 'A
    ↳ dictionary containing the status of the operation and the most recent health and mood
    ↳ records.'}}}}}
```

Corresponding preferences related to the tools you retrieve:

```
[User's preference related to ``get_recent_health_and_mood_summary`` and
    ↳ ``get_user_recent_workout_records``]
```

ASSISTANT:

Thought: To provide a personalized and proactive fruit recommendation for user, I need to

- ↳ consider his dietary habits, health metrics, and recent activities. By understanding his
- ↳ current health and mood status, as well as his recent workout records, I can tailor the fruit
- ↳ suggestions to align with his wellness priorities and lifestyle.

Action: get_recent_health_and_mood_summary

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent health and mood records]

ASSISTANT:

Thought: I will attempt to retrieve the recent workout records in order to recommend fruits that

- ↳ are appropriate for the user.

Action: get_user_recent_workout_records

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent workout records]

ASSISTANT:

Thought: I now know the final answer.

Final Answer:

Based on your current health and mood, which are [user's health status] after a day with moderate

- ↳ activity. His recent workout included [recent workout categories], focusing on [health
- ↳ concern and psychological requirements]. Considering the diet preference of you like
- ↳ [dietary preferences] and wellness priorities like [user's health goal], I recommend the
- ↳ following fruits that align with your wellness goals:

1. **[Fruit_1]**: [The reasons for recommending fruit 1].

2. **[Fruit_2]**: [The reasons for recommending fruit 2].

These fruits not only fit your diet preference of [dietary preferences] but also support your goals of
→ [user's health goal]. Enjoy them as part of your balanced, wellness-oriented diet! If you are
→ interested in other fruits, let me know!

Prompt 10: E-ReAct (Tool-Retrieval)

I need you to evaluate whether the solution provided by my artificial intelligence assistant completes
→ user instructions and meets user preferences.

Evaluation Metrics:

The analysis should assess the assistant's performance based on the following criteria:

- **Procedure**: Whether the process is correct and a complete and accurate final response was
→ provided.
- **Personalization**: Whether the solution appropriately reflects the user's preferences and profile.
- **Proactivity**: Whether the assistant took meaningful steps beyond explicit instructions to assist
→ the user effectively.

Evaluation Guidelines:

1. **Procedure Analysis**:

Assess the AI assistant's **entire solution process** (including tool usage, logic, and responses) and
→ its **final output**. Determine if the solution:

- Accurately completes the task based on the user's instructions.
- Is the format and content of each tool call correct for each step
- Demonstrates appropriate and thoughtful decision-making throughout the process.
- Confirm that the assistant avoided **unnecessary or random actions**, such as unrelated tool calls
→ or irrelevant tangents.
- Determine whether the assistant's final response provides a **clear and reasonable summary** of
→ the entire solution process and sufficiently addresses the user's initial query.

Score Criteria:

5 Points (Excellent):

Fully resolves the issue, no errors/omissions, rigorous logic, precise and relevant tool usage, concise
→ yet comprehensive content.

4 Points (Good):

Mostly correct with minor errors/omissions, tools generally accurate, relevant but lacks depth.

3 Points (Adequate):

Partial accuracy, noticeable omissions/errors, occasional tool misuse, content relevant but unclear.

2 Points (Poor):

Fails key requirements, major errors, frequent tool misuse, vague/irrelevant content.

1 Points (Unacceptable):

Irrelevant/severe errors, incorrect tool usage, content lacks value.

0 Points (Invalid):

No meaningful response, completely irrelevant/harmful, tools unused or grossly misapplied, zero

↪ value or violates guidelines.

2. ****Personalization Assessment****:

Evaluate whether the assistant meets the ****personal**** criterion:

- Did the assistant consider the user's specific preferences, profile details, and context when calling
↪ tools or generating the response?
- Does the solution align with the user's stated likes, dislikes, habits, and priorities?

Score Criteria:

5 Points (Excellent):

Perfectly aligns with user preferences/context, integrates unique details flawlessly, thoughtful and

↪ precise customization.

4 Points (Good):

Closely aligned with core needs, covers key preferences, minor gaps in depth or specificity.

3 Points (Adequate):

Basic personalization, misses critical details (e.g., specific scenarios), requires stronger tailoring.

2 Points (Poor):

Generic response, ignores major preferences, lacks relevance to user's unique context.

1 Points (Unacceptable):

No user context considered, entirely generic/irrelevant, provides no practical value.

0 Points (Invalid):

No personalization, irrelevant/harmful content, ignores user context, or violates guidelines.

3. ****Proactivity Behavior Assessment****:

Evaluate whether the assistant meets the ****proactive**** criterion:

- Did the assistant anticipate additional needs, explore further helpful insights, or propose actions
↪ that go beyond the user's explicit instructions?
- Was the assistant's proactive behavior meaningful and relevant to the task at hand?

Score Criteria:

5 Points (Excellent):

Anticipates ****unstated needs****, provides critical insights (e.g., long-term implications, unstated

↪ needs), and fully covers all proactive keypoints and dimensions with high relevance.

4 Points (Good):

Identifies ****most**** key proactive needs and keypoints (e.g., risk warnings, alternative solutions,

↪ helpful advices), adding clear value with minor gaps in depth.

3 Points (Adequate):

Addresses ****some**** proactive points (e.g., basic follow-up steps), but misses critical opportunities

↪ for deeper optimization.

2 Points (Poor):

Minimal proactivity; suggestions are superficial, incomplete, or lack relevance (e.g., generic tips

↪ unrelated to the context).

1 Point (Unacceptable):

Strictly passive; only fulfills explicit instructions with ****zero**** added insights, actions, or

↪ anticipation of needs.

0 Points (Invalid):

No proactive effort, provides harmful/irrelevant suggestions (e.g., dangerous advice), or completely

↪ misinterprets the task.

****Analysis Format:****

Your analysis should follow this JSON structure:

```
```json
{output_format}
```
```

****Evaluation Input:****

The evaluation will consider the following inputs:

1. ****User Query****:
{query}
2. ****User Profile****:
{profile}
3. ****Personal LLM Assistant Solution****:
{output}

****Analysis and Results:****

Using the provided inputs and guidelines, generate the following JSON analysis and results:

1. Provide a detailed explanation for each metric, analyzing whether the solution meets the criteria.
→ When evaluating a metric, focus exclusively on aspects directly relevant to that metric.
2. Assign a score (0~5) for each metric.
3. Ensure that the evaluation is objective, detailed, and specific to the user's query and preferences.

****Note:****

The format for the model's response is: `{{"role": "assistant", "content": "", "tool_calls": [{{"name": "",
→ "arguments": ""}}]}}`

A successful tool call must include feedback from the tool, rather than the model inferring or

- answering its own questions. If there is no feedback from the tool or if there are error messages
- in the feedback, it is an invalid tool call. The format for tool feedback is: `{{"role": "tool",
→ "content": ""}}`

If there is the user feedback in the conversation, check if there is possible tool-invoking format error or

- invalid tool utilizing. If exists, it indicates the error or unnecessary tool calling.

Prompt 11: key-point-based LLM Evaluation 的评测 Prompt

```
{
  "Procedure": {
```

```

"Keypoints for Procedure": {
  "Completeness": {
    "analysis": "The assistant successfully identified the user's preferences for jazz and
    ↪ classical music, selected a track ('So What' by Miles Davis) from the user's
    ↪ frequently played tracks, and set the volume to the user's preferred level of
    ↪ 40%. The assistant also provided a final answer summarizing the task
    ↪ completion. However, there was an unnecessary user prompt due to the
    ↪ assistant's failure to finalize the response earlier, which required the user to
    ↪ intervene and request a proper conclusion.",
    "score": 1
  },
  "Avoid Unnecessary Action": {
    "analysis": "The assistant made an unnecessary tool call after the music was already
    ↪ playing ('tool_calls': [{'name': '', 'arguments': '{}'}]). This action did not
    ↪ contribute to the task and caused confusion. Additionally, the assistant failed
    ↪ to finalize the response initially, leading to unnecessary user feedback.",
    "score": 1
  },
  "Call the tool accurate": {
    "analysis": "The tool calls were mostly accurate. The assistant correctly used
    ↪ 'get_music_list_in_favorites' to retrieve the user's favorite tracks and
    ↪ 'play_music' to play the selected track at the specified volume. However, the
    ↪ unnecessary tool call ('tool_calls': [{'name': '', 'arguments': '{}'}]) after
    ↪ the music was already playing detracts from the accuracy of tool usage.",
    "score": 1
  },
  "Summary the query clearly and comprehensive": {
    "analysis": "The assistant provided a clear and concise final answer summarizing the
    ↪ task ('So What' by Miles Davis is now playing at a volume level of 40. Enjoy
    ↪ your relaxing morning!'). However, this summary was delayed and only
    ↪ provided after the user prompted the assistant to finalize the response.",
    "score": 1
  }
},
"Final Assessment": {
  "analysis": "The assistant partially completed the task but exhibited issues with unnecessary
  ↪ actions, delayed finalization, and redundant tool calls. While the task was
  ↪ eventually completed, these errors detract from the overall performance.",
  "score": 3
}
},
"Personalization": {
  "Keypoints for Personalization": {
    "Consider user's preferred genre or artists for music ('MusicPreferences') when using
    ↪ 'get_music_list_in_favorites'.": {
      "analysis": "The assistant correctly considered the user's preferred genres (jazz and
      ↪ classical) and selected a track ('So What' by Miles Davis) from the user's
      ↪ frequently played tracks. This aligns with the user's preferences.",
      "score": 2
    },
    "Consider user's usual music preferences and listening habits ('ListeningHabits') based on
    ↪ the time of day.": {

```

```

        "analysis": "The assistant acknowledged the user's preference for jazz and classical
            ↳ music for relaxation and selected a track suitable for the morning routine. This
            ↳ demonstrates consideration of the user's listening habits.",
        "score": 2
    },
    "Consider user's preferred volume level ('PreferredVolumeLevel') for music when using
        ↳ 'play__music'.": {
        "analysis": "The assistant set the volume to 40%, which aligns with the user's preferred
            ↳ volume level for immersive listening.",
        "score": 2
    }
},
"Final Assessment": {
    "analysis": "The assistant demonstrated excellent personalization by considering the user's
        ↳ preferred genres, listening habits, and volume preferences. The solution was
        ↳ well-tailored to the user's profile.",
    "score": 5
}
},
"Proactivity": {
    "Keypoints for Proactivity": {
        "Provide other music suggestions besides the music you recommend and play.": {
            "analysis": "The assistant did not provide additional music suggestions beyond the
                ↳ selected track ('So What' by Miles Davis). This was a missed opportunity to
                ↳ enhance the user's experience.",
            "score": 0
        },
        "Ask if user prefers a new recommendation of different music category": {
            "analysis": "The assistant did not ask the user if they would like a recommendation
                ↳ from a different music category. This was another missed opportunity to
                ↳ proactively engage with the user.",
            "score": 0
        },
        "Consider the time of the day when you recommend music and determine volume.": {
            "analysis": "The assistant considered the time of day (morning) and selected a relaxing
                ↳ jazz track while setting the volume to 40%, which aligns with the user's
                ↳ preferences for immersive listening during this time.",
            "score": 2
        }
    },
    "Final Assessment": {
        "Analysis": "The assistant demonstrated limited proactivity. While it considered the time
            ↳ of day when selecting music and setting the volume, it missed opportunities to
            ↳ provide additional suggestions or ask about alternative preferences.",
        "score": 2
    }
}
}

```

Example 12: 评测结果示例